MLR780 Air Conditioning and Refrigeration 2021

Assignment 9
Vapour Compression Cycle Analysis

Instructions

- This individual assignment is to be submitted online before midnight on **4 July 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment9-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment9-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Outcome

This assignment is designed to enable you to demonstrate your ability to analyse a vapour compression cycle. A multitude number of cycle architectures exist. Only a single arbitrary architecture is considered in this assignment. The heat transfer behaviour during phase change is complex and is described by a specialist field which falls beyond the scope of this module. Simplifications are made in this regard for purposes of this assignment.

Introduction

Consider a vapour compression cycle operating with R410a. A variable speed compressor is used with an inlet at state 1 which is carefully controlled by a constant enthalpy expansion device such that 5°C superheat is always maintained at the compressor inlet. (IE the temperature is 5°C higher than the saturation evaporator temperature). The compressor has a 4.5% clearance and a volumetric displacement rate which is directly proportional to the compressor motor speed. Assume the compressor is isentropic (in reality it will not be).

A single condenser is used which is air cooled and which has an inlet at state 2 and an outlet at state 3. Assume the condenser fully condenses the refrigerant to a saturated liquid phase (quality = 0). The heat transfer rate in the condenser, q_c, has been simplified for purposes of this assignment to be only dependent on the condensing temperature (saturated temperature), t_c, and the air inlet temperature, $t_{ac,in}$:

$$q_c = F\left(t_c - t_{ac,in}\right)$$

Let F = 7500 W/K and $t_{ac,in}$ =35°C. (Note that in reality F is a function of the refrigerant mass flux, vapour quality, and the condensing temperature.)

A single evaporator is used to chill water for the cooling and dehumidification coil used in a VAV system in an office building. The refrigerant at the inlet is at state 4 and at the outlet is at state 1. The heat transfer rate in evaporator is expressed for purposes of this assignment as follows:

$$q_e = G\left(t_{we,out} - t_e\right)$$

Here $t_{we,out}$ is the **exit** water temperature **out of** the evaporator, t_e is the saturation evaporation temperature, and G [W/K] can be expressed in terms of the water mass flow rate, $\dot{m}_{we}$ [kg/s] as:

$$G = 7000\left(0.57\dot{m}_{we}^{0.8} + 0.046\left(t_{we,out} - t_e\right)\right)$$

Let $\dot{m}_{we}$ = 1.4 kg/s. (Note that in reality G is also a function of the refrigerant mass flux, vapour quality, and the evaporation temperature, but has been simplified for purposes of this assignment.)

The cooling coil has different cooling load requirements, q_{coil} [W], depending on the cooling need in the office building. For purposes of this assignment this is given as follows depending on the hour in the day, $\theta \in [0,24] \epsilon \mathbb{N}$:

$$q_{coil} = 13000\left[cos\left(\frac{\theta + 9}{12}\pi\right) + 3\right]$$

For purposes of this assignment, the air and water flow rates through the condenser and the evaporator are fixed. To prevent freezing of the water in the evaporator, the evaporation temperate must always be above 2°C while the specifications of the cooling coil requires the water inlet temperature to always be 10°C (±0.1°C).

Use the clearance volume efficiency to perform the following calculations:

1. At hour zero ($\theta = 0$), when the cooling coil requires 29.8 kW, analyse the vapour compression cycle if the compressor is running at a volumetric displacement rate of 2 L/s. Consider these two cases:
    a. t_e=10°C and t_c=40°C (not the correct balancing temperature)
    b. Fully balanced operating point (converged t_e and t_c)

    For each of the above. tabulate all thermodynamic properties for states 1 to 4 (temperature, pressure, enthalpy and entropy), as well as other important quantities such as the clearance volumetric efficiency, refrigerant mass flow rate, condenser heat transfer rate, COP, and total compressor power requirements etc. Clearly define all definitions and equations

used. Indicate which mathematical/ solution method or approach was used.

2. Analyse the balanced operational points for all the hours from 8 to 18 and determine the required compressor displacement rate to satisfy the cooling coil water inlet and evaporating temperatures. Support your answer with a suitable discussion and calculations with tabulated results for hour 12 (similar to the tabulations in question 1).
3. Plot the required compressor displacement rate, power requirement and the overall COP for the hours between 08 and 18.
4. Assuming the expressions for F and G remains unchanged (in reality F and G are also influenced by the refrigerant type), could a different selection of refrigerant result in a better system? Repeat question 2, but only at hour 12 for the following refrigerants:
    a. R134a
    b. R717 (ammonia)
    c. R744 (carbon dioxide)

    Only consider options that will still result in an actual standard vapour compression cycle. Also report on the maximum system pressure as well as the refrigerant's critical point temperature. Plot the COP, t_e, t_c, and maximum system pressure on bar charts to help you discuss your results.

INDICATE HOW MANY HOURS YOU SPENT ON THIS ASSIGNMENT